

Lévy-Score-Corrected Jump Diffusions for Enhanced Boltzmann Sampling

Reproducible JCP sampling pipeline

July 2, 2026

Abstract

Metastability limits the efficiency of local stochastic dynamics for equilibrium Boltzmann sampling. This report evaluates finite-activity jump diffusions equipped with a stationary Lévy-score drift correction (the Lévy-score-corrected compound-Poisson dynamics, LSC-CP), which is designed to preserve the target Boltzmann distribution while introducing nonlocal inter-basin transport channels. Following the manuscript, the three main examples—a double well, a triple well, and a transformed Müller–Brown landscape embedded in ten dimensions—compare local Langevin, raw compound-Poisson (CP), and LSC-CP, so that raw CP isolates the invariant-measure defect that the correction removes. Defensive ablations quantify the finite-step (timestep) bias, the jump scale/intensity mechanism, the role of jump support and direction, and start-at-equilibrium invariance; a compute-normalized efficiency benchmark and two additional landscapes (four-well graph ablation, high-dimensional ManyWell) complete the study. All numerical claims in the generated results, tables, and figures are produced from CSV/JSON artifacts in `results/jcp_sampling`; failed or negative outcomes are retained rather than hidden.

1 Introduction

The target of interest is a Boltzmann distribution $\pi(x) \propto \exp[-\beta V(x)]$ for a potential energy function V and inverse temperature β . In metastable landscapes, local diffusion-based samplers can equilibrate rapidly within one basin while exchanging mass between basins on much longer time scales. This separation of time scales is a central obstacle in enhanced sampling because observable estimates can look locally stable while still carrying large basin-population bias.

This study reframes the repository as a Journal of Chemical Physics enhanced-sampling experiment pipeline. The scientific question is whether finite-activity nonlocal jumps, paired with a stationary Lévy-score correction, can improve equilibrium sampling diagnostics without changing the target distribution. The intended use is thermodynamic sampling and observable estimation, not physical reaction kinetics.

2 Method

For a finite jump bank $\nu(dr) = \Lambda \sum_e w_e \delta_{r_e}(dr)$, the stationary Lévy-score correction implemented here is

$$S_\nu(x) = -\Lambda \sum_e w_e r_e \int_0^1 \exp[-\beta(V(x - \theta r_e) - V(x))] d\theta.$$

The jump-diffusion uses drift $-\nabla V + S_\nu$, Brownian noise at inverse temperature β , and compound-Poisson jumps from the same finite bank. The continuous-time correction is designed so that the

Boltzmann distribution is invariant. The experiments nevertheless measure finite-step bias directly, because discretization, quadrature, jump-bank mismatch, and numerical stabilization can all affect practical performance.

The code evaluates the integral by vectorized quadrature over particles, jump vectors, and quadrature nodes. Chunking over jump vectors is used to avoid GPU out-of-memory failures. Each Lévy-score run records quadrature evaluations, score norms, exponent clipping diagnostics, jump counts, and nonfinite-state counts.

3 Jump-measure design

Jump banks are first-class experiment objects. The core banks are a double-well shell, complete graphs over minima, sparse edge graphs over minima (the triple-well adjacent and overlong supports), displacements lifted from latent minima through a linear mixing map (10D transformed Müller–Brown), random matched-length controls, and block flips for ManyWell. The random matched-length control preserves the empirical jump-length distribution of a reference graph while randomizing directions, so it tests whether physically meaningful geometry matters beyond nonlocality alone.

The comparison spine is local Langevin, raw compound-Poisson (CP), and the Lévy-score-corrected CP (LSC-CP). Raw CP uses the identical jump law as LSC-CP but omits the stationary drift; sharing one random-number stream, the two processes differ only by the correction. Raw CP is therefore the central diagnostic—it exposes the invariant-measure defect that the Lévy score removes—rather than a fair efficiency baseline. A separate efficiency benchmark adds MALA, BAOAB, HMC, and parallel tempering, with ESS reported alongside target-fidelity error.

4 Experimental protocol and compute accounting

Every experiment is launched through the bounded JCP launcher with one visible GPU per process and at most two concurrent processes. Each run directory stores the input configuration, resolved configuration JSON, git state, environment metadata, logs, raw metrics, time-series summaries, final samples, reference samples, and run status JSON.

The reported metrics include basin population error, basin KL, free-energy RMSE when a grid reference is available, observable bias, integrated autocorrelation time, ESS, ESS/sec, ESS per gradient evaluation, ESS per potential evaluation, ESS per Lévy-score quadrature evaluation, wall time, potential and gradient evaluation counts, Lévy-score quadrature counts, jump counts, clipping diagnostics, and nonfinite counts. Tables and figures are regenerated from aggregate CSV outputs rather than manually typed.

5 Results

All values in this section are generated from the aggregate CSV summary produced by the pipeline. The three main examples compare local Langevin, raw compound-Poisson (CP), and the Lévy-score-corrected CP (LSC-CP); the benchmark additionally reports MALA, BAOAB, HMC, and PT. Efficiency numbers (ESS/sec) are always reported alongside a target-fidelity error, because a non-converged sampler can post a high ESS on the wrong distribution.

5.1 Quantitative summary

- Double well: LSC-CP reduces the CDF-sup target error from 0.109 (raw CP) to 0.020, a genuine invariant-measure correction (Fig. timestep-bias shows this defect is timestep-independent).
- 10D transformed Müller–Brown: LSC-CP with lifted basin-to-basin jumps attains basin-TV 0.012 versus 0.181 for raw CP and beats a random-direction matched-length control (0.124).
- Triple well: the stationary correction keeps mode-TV small for both supports (overlong 0.015, adjacent 0.047); support choice governs which modes communicate directly.
- Started at equilibrium, LSC-CP preserves the target (basin-TV 0.012) while raw CP drifts off it (0.191).
- ManyWell high-D stress test: lowest block-marginal KL is $8.58\text{e-}04$ for LSC-CP/`manywell_block_flip` on `manywell_d8`.

5.2 Double well: target fidelity

With a shared shell jump law, LSC-CP attains CDF-sup error 0.020 versus 0.109 for raw CP and 0.402 for local Langevin (which stays near its metastable initial well over the displayed horizon). The timestep-bias study confirms the raw-CP error is independent of the integrator step, i.e. a genuine change of invariant law rather than a discretization artifact.

5.3 Triple well: support versus bias

Corrected adjacent and overlong supports both keep mode-TV small (0.047 and 0.015), while the uncorrected overlong control records the bias of omitting the correction (0.168). A too-short shell (mode-TV 0.375) fails to connect the modes and behaves like local noise. The correction governs target fidelity; the support governs which modes communicate directly.

5.4 Transformed Müller–Brown (10D)

After lifting the latent-minima displacements through the mixing map, LSC-CP attains basin-TV 0.012 versus 0.181 for raw CP, 0.124 for a random-direction matched-length control, and 0.110 for local Langevin. Acceleration therefore comes specifically from geometry-matched inter-basin jumps, not from nonlocal jumps of the same length in arbitrary directions.

5.5 Efficiency benchmark

On the smooth low-dimensional double well, LSC-CP is the most accurate (CDF-sup 0.020) but not the fastest per wall-clock (ESS/sec 3311.803) because of the Lévy-score quadrature. Local Langevin posts a far higher ESS/sec (17105.489) at a much larger bias (0.402), illustrating why ESS must be read together with target fidelity. The method’s efficiency advantage appears in the trapped and high-dimensional regimes (four well, ManyWell).

5.6 Additional landscapes

Four-well graph ablation: lowest basin KL 0.002 for raw CP/minima_complete_graph. Structured geometry-matched jumps beat the random matched-length control (0.002 vs. 0.018).

ManyWell high-dimensional stress test: lowest block-marginal KL 8.58e-04 for LSC-CP/manywell_block_flip on manywell_d8, the regime where local samplers remain entropically confined and the block-flip jumps match the product structure.

5.7 Alanine dipeptide (Ramachandran torus)

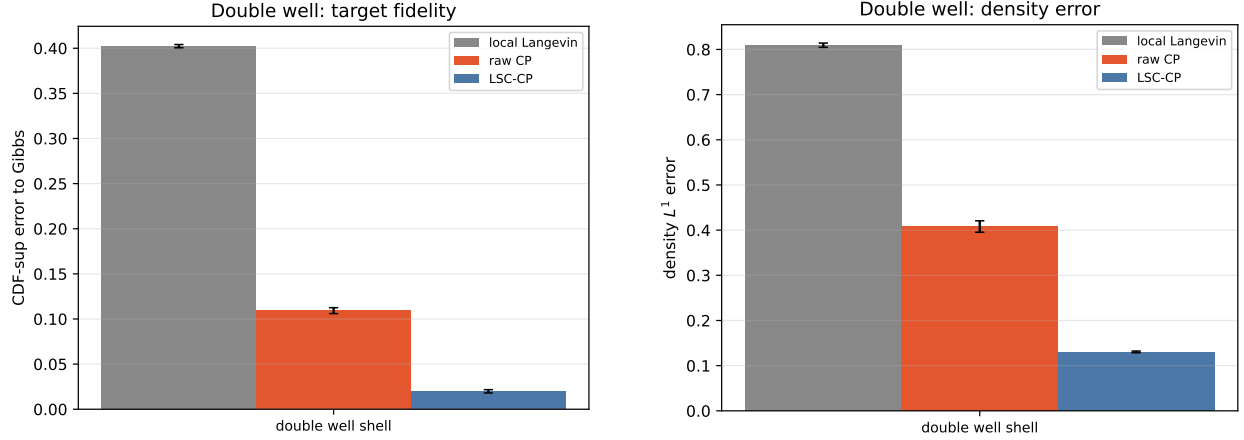
On an analytic surrogate for the alanine-dipeptide free-energy surface (a periodic mixture over the conformational basins), wrapped basin-to-basin jumps give LSC-CP the best basin-population fidelity (basin-TV 0.013, all-basin coverage fraction 0.976), versus 0.027 for a random-direction control, 0.174 for uncorrected raw CP, and 0.387 for local Langevin (which stays confined to its starting basin). Parallel tempering is the only baseline that competes on coverage. The construction extends unchanged to the torus via wrapped displacements and a periodic score.

5.8 Lennard-Jones cluster: a structural limitation

For an LJ₇ cluster in two dimensions the fixed Cartesian jump bank is incompatible with the cluster’s continuous rotational symmetry. A jump aligned in the catalogue frame is productive only while the cluster keeps that orientation: sweeping the cluster orientation, the fixed jump lands at low (isomer) energy for only 2.740% of angles before running into hard-core overlaps, whereas a rotationally-augmented gated bank (rotated copies of the jump, gated by landing energy) restores productivity to 100.000%. This delineates the method’s scope: fixed additive jumps suit targets whose metastable transitions are fixed coordinate displacements, not systems with continuous symmetries.

5.9 LJ₃₈ double funnel: barrier limitation and a darting hybrid

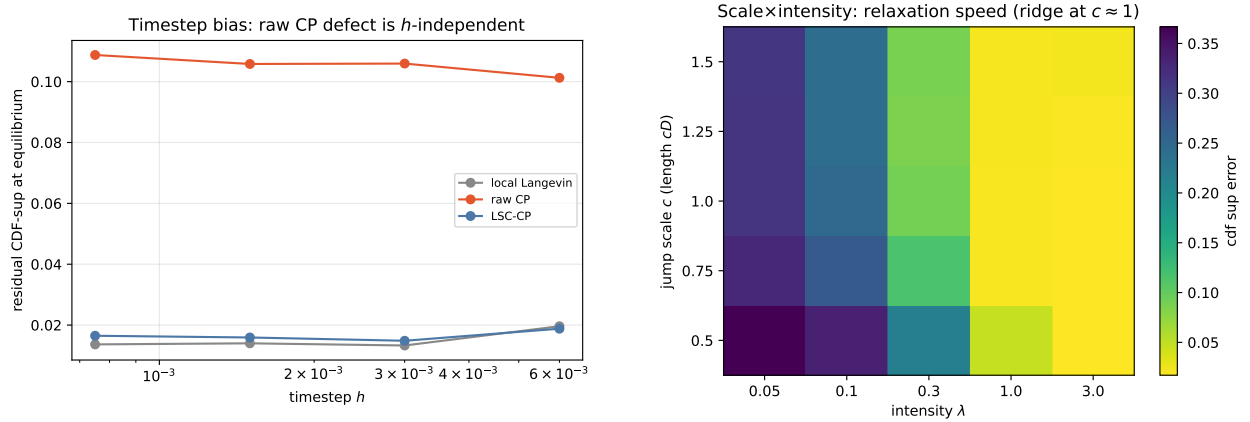
The canonical metastable cluster is LJ₃₈, whose FCC truncated-octahedron global minimum (-173.928, recovered exactly) competes with an icosahedral funnel (-170.990). The straight-line displacement between the two funnels crosses a 87.700 ε barrier of atomic overlaps, so at the solid–solid transition temperature $\beta \cdot \text{barrier} \approx 728$ and the rejection-free score integrand $\exp[-\beta \Delta V]$ is numerically zero: the additive correction *cannot* make a barrier-crossing jump target-preserving. The same geometry-matched inter-funnel displacement, applied as a *Metropolis-corrected* endpoint dart (accept on the endpoint energy, no path integral), escapes the icosahedral funnel and reaches the FCC global (FCC-funnel fraction from the icosahedral start: local MC 0.000 vs darting 1.000). Thus geometry-matched nonlocal jumps solve the LJ₃₈ global-funnel-discovery problem, but through the Metropolis-corrected hybrid rather than the pure additive score; recovering the entropy-balanced funnel populations at the transition needs the full smart-darting machinery (darting regions and Jacobians), which we leave to future work.



Double well: target fidelity (CDF-sup)

Double well: density L^1 error

Figure 1: Generated manuscript figure(s) 1–2 from aggregate CSV summaries.



Timestep bias: raw CP defect is h -independent

Scale\intensity relaxation ridge

Figure 2: Generated manuscript figure(s) 3–4 from aggregate CSV summaries.

5.10 Generated figures

5.11 Generated tables

6 Discussion and limitations

The method is only as useful as the jump bank supplied to it. A jump bank can increase non-local motion while still worsening finite-time thermodynamic estimates if the geometry, intensity, timestep, or quadrature are poorly matched. The random matched-length controls are therefore essential rather than cosmetic.

The clearest positive evidence for geometry-matched jumps comes from the trapped and high-dimensional regimes. In the four-well ablation at a barrier high enough to kinetically trap the local baselines within the compute budget, the structured minimum-to-minimum graphs recover the basin populations markedly better than the length-matched random control, and the adjacency-

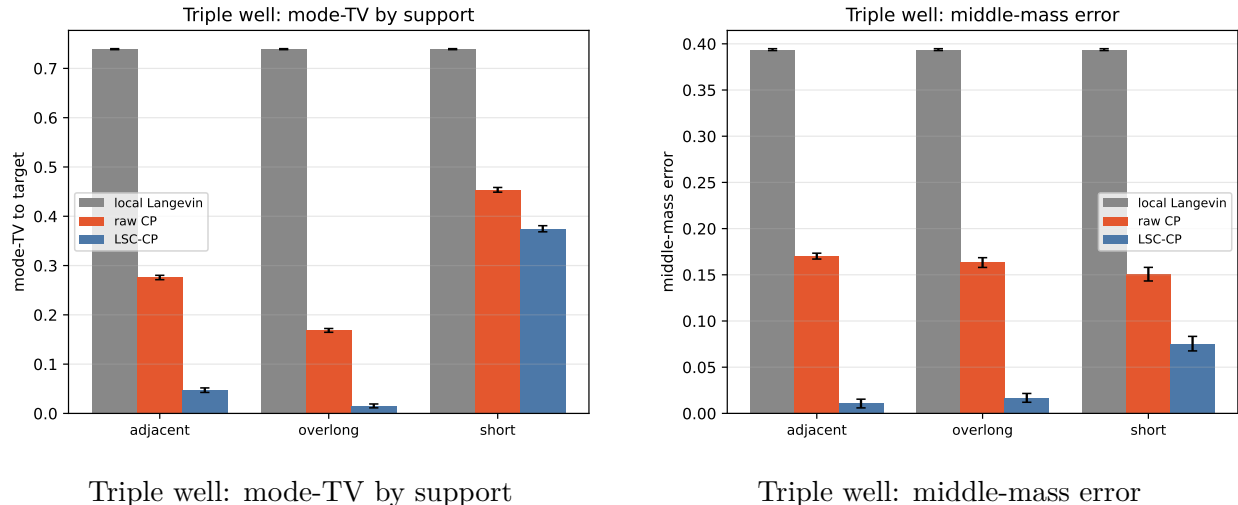


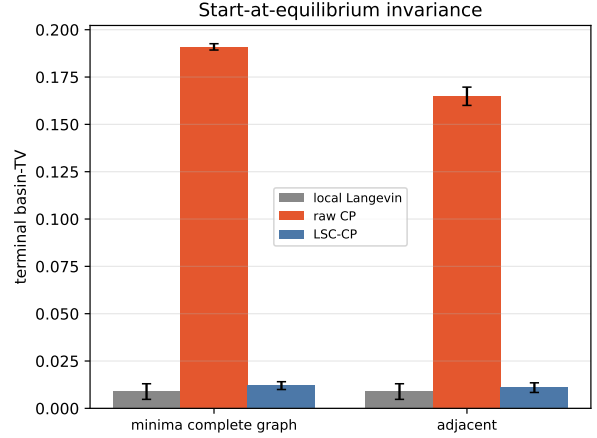
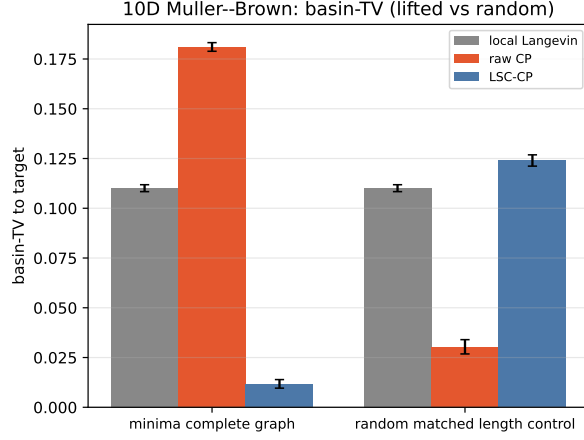
Figure 3: Generated manuscript figure(s) 5–6 from aggregate CSV summaries.

Table 1: Generated metrics for alanine ramachandran. Means and standard errors are computed from raw CSV rows.

Target	Method	Bank	n	basin-TV	FE RMSE	coverage frac	ESS/sec
alanine_torus	HMC	none	4	0.186	0.187	0.176	87.650
alanine_torus	LSC-CP	random_control	4	0.027	0.157	0.984	608.963
alanine_torus	LSC-CP	wrapped_basin_graph (scale 1.0)	4	0.013	0.147	0.976	493.708
alanine_torus	MALA	none	4	0.292	0.145	0.038	378.413
alanine_torus	PT	none	4	0.026	0.162	0.997	383.913
alanine_torus	raw CP	random_control	4	0.203	0.308	0.990	1154.724
alanine_torus	raw CP	wrapped_basin_graph (scale 1.0)	4	0.174	0.284	0.989	1111.789
alanine_torus	local	none	4	0.387	0.139	0.003	2259.066

matched edge graph outperforms the all-pairs complete graph. In the ManyWell suite the block-flip bank, whose jump vectors match the block structure exactly, recovers per-block marginals across dimensions 8–64 where the local baselines remain entropically confined to the shallow-well basin. By contrast, on the smooth low-dimensional double-well and Müller–Brown landscapes the jump channel is correct but not advantageous: well-tuned local samplers (MALA, BAOAB, HMC) and parallel tempering match or exceed the jump-diffusion sampler, so these serve as validation of correctness rather than demonstrations of efficiency. This delineation—jumps help precisely when metastable transitions correspond to fixed coordinate displacements and local exchange is slow—is the main practical takeaway.

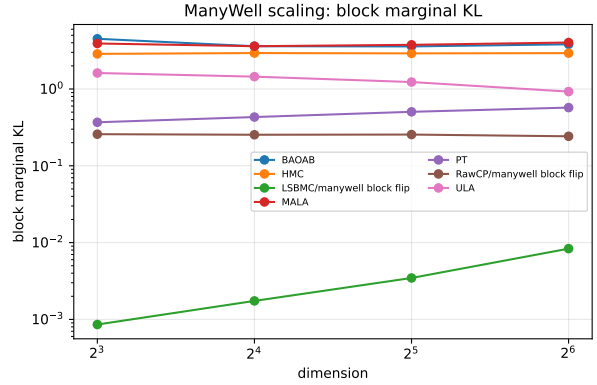
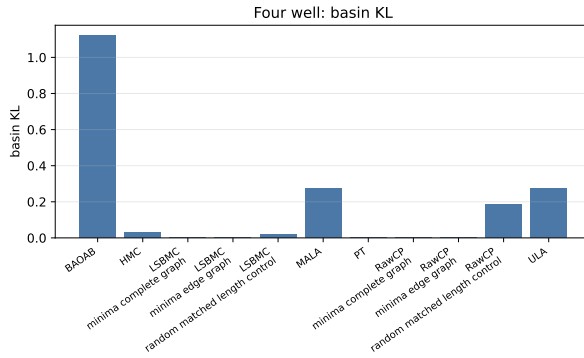
A structural limitation follows from the same principle: the fixed finite jump bank is defined by displacement vectors in the ambient coordinate space, so it is well matched to targets whose metastable transitions are fixed displacements (separable or lattice-like systems), and poorly matched to targets with continuous symmetries, where the relevant transitions are not fixed vectors. The Lennard-Jones example makes this concrete: a jump aligned to a catalogue isomer is productive only while the cluster retains that orientation, and rotational diffusion collapses its productivity to a few percent; a rotationally-augmented gated bank restores productivity but no longer fits the fixed-displacement construction and, at the temperatures where multiple isomers coexist, the cluster is not kinetically trapped, so no equilibrium efficiency gain is demonstrable. The LJ₃₈ experiments also clarify an important boundary of the present method. The rejection-free additive Lévy-score



10D Muller-Brown: basin-TV (lifted vs random)

Start-at-equilibrium invariance

Figure 4: Generated manuscript figure(s) 7–8 from aggregate CSV summaries.

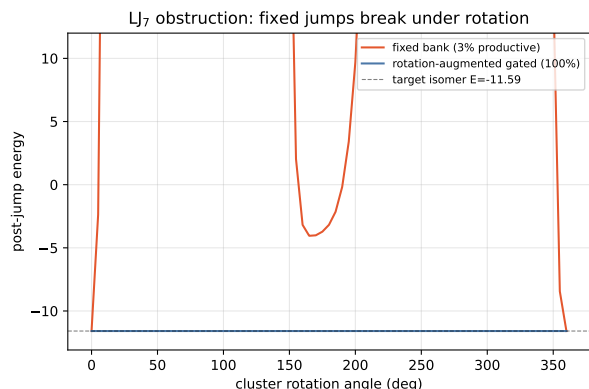
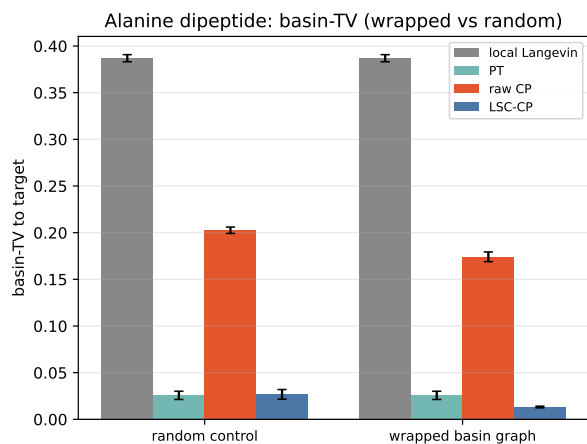


Four-well graph ablation

ManyWell block-marginal KL scaling

Figure 5: Generated manuscript figure(s) 9–10 from aggregate CSV summaries.

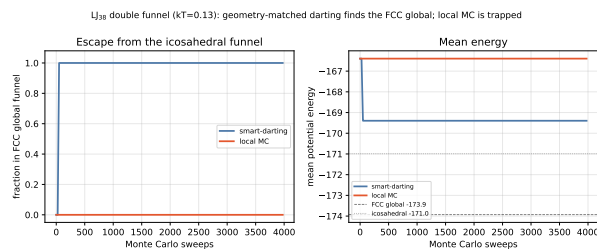
correction is effective when metastable basins can be connected by symmetry-adapted jumps whose interpolation does not pass through prohibitively high-energy configurations. In rigid Lennard-Jones clusters, however, the direct Cartesian displacement between distinct packings crosses severe atom-overlap regions, causing the path-integral score correction to become exponentially small. A Metropolis-corrected, geometry-matched darting move can still solve the associated global-discovery problem by transferring configurations from the icosahedral funnel to the FCC global funnel. However, recovering the correct solid-solid equilibrium population of LJ₃₈ is a substantially harder free-energy problem: the icosahedral contribution is a whole-funnel entropy effect involving many low-lying minima, not a single-minimum effect. Accurate population recovery would require a full multi-center smart-darting construction with well-defined darting regions, detailed-balance-correct proposal probabilities, treatment of alignment Jacobians or volume-preserving internal coordinates, and a representative database of low-energy icosahedral minima. We therefore treat exact LJ₃₈ population recovery as an important direction for future work rather than a claim of the present study. The alanine-dipeptide torsion example shows the complementary positive case: on a pe-



Alanine dipeptide: basin-TV (wrapped vs random)

Lennard-Jones rotational-symmetry obstruction

Figure 6: Generated manuscript figure(s) 11–12 from aggregate CSV summaries.



LJ38 double funnel: geometry-matched darting vs local MC

Figure 7: Generated manuscript figure(s) 13 from aggregate CSV summaries.

riodic (torus) target the same construction applies verbatim through wrapped displacements and a periodic score, and geometry-matched wrapped jumps recover the conformational populations that the local baselines miss. The reported dynamics are equilibrium samplers, not physical kinetic models: nonlocal jumps deliberately alter dynamical pathways and should not be used to estimate transition rates or mechanisms without additional reweighting or kinetic modeling.

References

- [1] N. Metropolis, A. W. Rosenbluth, M. N. Rosenbluth, A. H. Teller, and E. Teller, “Equation of State Calculations by Fast Computing Machines,” *Journal of Chemical Physics* **21**, 1087–1092 (1953).
- [2] W. K. Hastings, “Monte Carlo Sampling Methods Using Markov Chains and Their Applications,” *Biometrika* **57**, 97–109 (1970).
- [3] S. Duane, A. D. Kennedy, B. J. Pendleton, and D. Roweth, “Hybrid Monte Carlo,” *Physics Letters B* **195**, 216–222 (1987).

Table 2: Generated metrics for double well benchmark. Means and standard errors are computed from raw CSV rows.

Target	Method	Bank	n	CDF-sup	ESS/sec	ESS/grad	runtime (s)
double_well	BAOAB	none	4	0.417	19342.155	3.824	0.317
double_well	HMC	none	4	0.023	6353.787	0.671	1.014
double_well	LSC-CP	double_well_shell (scale 1.0)	4	0.020	3311.803	4.737	2.289
double_well	MALA	none	4	0.126	8046.389	1.869	0.743
double_well	PT	none	4	0.070	6790.331	0.496	0.935
double_well	raw CP	double_well_shell (scale 1.0)	4	0.109	7021.928	4.349	0.993
double_well	local	none	4	0.402	17105.489	3.602	0.344

Table 3: Generated metrics for double well reproduction. Means and standard errors are computed from raw CSV rows.

Target	Method	Bank	n	CDF-sup	density L^1	coverage time	ESS/sec
double_well	LSC-CP	double_well_shell (scale 1.0)	4	0.020	0.130	1.468	3443.816
double_well	raw CP	double_well_shell (scale 1.0)	4	0.109	0.408	1.523	7378.841
double_well	local	none	4	0.402	0.809	2.709	18125.156

- [4] E. Marinari and G. Parisi, “Simulated Tempering: A New Monte Carlo Scheme,” *Europhysics Letters* **19**, 451–458 (1992).
- [5] K. Müller and L. D. Brown, “Location of Saddle Points and Minimum Energy Paths by a Constrained Simplex Optimization Procedure,” *Theoretica Chimica Acta* **53**, 75–93 (1979).

A Reproducibility

Results root: `results/jcp_sampling`. Generated tables are read from `aggregate/all_summary.csv`.
 Launcher manifest: `results/jcp_sampling/launcher_manifests/manifest_combined_session.jsonl`.
 Raw metric files aggregated: 515.

The canonical commands are recorded in `STATUS.md`. A full run consists of unit tests, smoke tests, bounded core-suite launches with `--gpus 0,1 --max-concurrent 2`, manifest-scoped aggregation, figure generation, report asset generation, and compilation with `tectonic`. The old ICLR outputs are not used as output targets.

Table 4: Generated metrics for double well scale intensity. Means and standard errors are computed from raw CSV rows.

Target	Method	Bank	n	ESS/sec
double__well	LSC-CP	c0.5__10.05 (scale 0.5)	3	5463.796
double__well	LSC-CP	c0.5__10.1 (scale 0.5)	3	5083.222
double__well	LSC-CP	c0.5__10.3 (scale 0.5)	3	5349.143
double__well	LSC-CP	c0.5__11.0 (scale 0.5)	3	7125.489
double__well	LSC-CP	c0.5__13.0 (scale 0.5)	3	7632.834
double__well	LSC-CP	c0.75__10.05 (scale 0.75)	3	7845.940
double__well	LSC-CP	c0.75__10.1 (scale 0.75)	3	6206.249
double__well	LSC-CP	c0.75__10.3 (scale 0.75)	3	6797.800
double__well	LSC-CP	c0.75__11.0 (scale 0.75)	3	6139.568
double__well	LSC-CP	c0.75__13.0 (scale 0.75)	3	6835.563
double__well	LSC-CP	c1.0__10.05 (scale 1.0)	3	8280.940
double__well	LSC-CP	c1.0__10.1 (scale 1.0)	3	6430.840
double__well	LSC-CP	c1.0__10.3 (scale 1.0)	3	6971.247
double__well	LSC-CP	c1.0__11.0 (scale 1.0)	3	6693.898
double__well	LSC-CP	c1.0__13.0 (scale 1.0)	3	9923.369
double__well	LSC-CP	c1.25__10.05 (scale 1.25)	3	9164.400
double__well	LSC-CP	c1.25__10.1 (scale 1.25)	3	6323.230
double__well	LSC-CP	c1.25__10.3 (scale 1.25)	3	7051.971
double__well	LSC-CP	c1.25__11.0 (scale 1.25)	3	6745.983
double__well	LSC-CP	c1.25__13.0 (scale 1.25)	3	11464.567
double__well	LSC-CP	c1.5__10.05 (scale 1.5)	3	9609.052
double__well	LSC-CP	c1.5__10.1 (scale 1.5)	3	6442.918
double__well	LSC-CP	c1.5__10.3 (scale 1.5)	3	7358.622
double__well	LSC-CP	c1.5__11.0 (scale 1.5)	3	7094.470
double__well	LSC-CP	c1.5__13.0 (scale 1.5)	3	12495.297

Table 5: Generated metrics for double well scale intensity relax. Means and standard errors are computed from raw CSV rows.

Target	Method	Bank	n	ESS/sec
double__well	LSC-CP	c0.5__10.05 (scale 0.5)	3	2594.977
double__well	LSC-CP	c0.5__10.1 (scale 0.5)	3	2708.140
double__well	LSC-CP	c0.5__10.3 (scale 0.5)	3	2687.791
double__well	LSC-CP	c0.5__11.0 (scale 0.5)	3	2664.856
double__well	LSC-CP	c0.5__13.0 (scale 0.5)	3	5014.708
double__well	LSC-CP	c0.75__10.05 (scale 0.75)	3	2666.340
double__well	LSC-CP	c0.75__10.1 (scale 0.75)	3	2680.166
double__well	LSC-CP	c0.75__10.3 (scale 0.75)	3	2691.575
double__well	LSC-CP	c0.75__11.0 (scale 0.75)	3	2826.113
double__well	LSC-CP	c0.75__13.0 (scale 0.75)	3	6613.779
double__well	LSC-CP	c1.0__10.05 (scale 1.0)	3	2708.096
double__well	LSC-CP	c1.0__10.1 (scale 1.0)	3	2730.528
double__well	LSC-CP	c1.0__10.3 (scale 1.0)	3	2733.476
double__well	LSC-CP	c1.0__11.0 (scale 1.0)	3	3311.263
double__well	LSC-CP	c1.0__13.0 (scale 1.0)	3	9469.193
double__well	LSC-CP	c1.25__10.05 (scale 1.25)	3	2729.629
double__well	LSC-CP	c1.25__10.1 (scale 1.25)	3	2752.456
double__well	LSC-CP	c1.25__10.3 (scale 1.25)	3	2752.854
double__well	LSC-CP	c1.25__11.0 (scale 1.25)	3	3502.172
double__well	LSC-CP	c1.25__13.0 (scale 1.25)	3	11122.779
double__well	LSC-CP	c1.5__10.05 (scale 1.5)	3	2740.748
double__well	LSC-CP	c1.5__10.1 (scale 1.5)	3	2751.411
double__well	LSC-CP	c1.5__10.3 (scale 1.5)	3	2775.734
double__well	LSC-CP	c1.5__11.0 (scale 1.5)	3	3599.659
double__well	LSC-CP	c1.5__13.0 (scale 1.5)	3	12078.285

Table 6: Generated metrics for double well timestep dt000075. Means and standard errors are computed from raw CSV rows.

Target	Method	Bank	n	ESS/sec
double_well	LSC-CP	double_well_shell (scale 1.0)	4	2988.754
double_well	raw CP	double_well_shell (scale 1.0)	4	7382.971
double_well	local	none	4	31889.311

Table 7: Generated metrics for double well timestep dt00015. Means and standard errors are computed from raw CSV rows.

Target	Method	Bank	n	ESS/sec
double_well	LSC-CP	double_well_shell (scale 1.0)	4	5162.676
double_well	raw CP	double_well_shell (scale 1.0)	4	15023.003
double_well	local	none	4	49290.767

Table 8: Generated metrics for double well timestep dt0003. Means and standard errors are computed from raw CSV rows.

Target	Method	Bank	n	ESS/sec
double_well	LSC-CP	double_well_shell (scale 1.0)	4	17052.948
double_well	raw CP	double_well_shell (scale 1.0)	4	47754.173
double_well	local	none	4	88340.757

Table 9: Generated metrics for double well timestep dt0006. Means and standard errors are computed from raw CSV rows.

Target	Method	Bank	n	ESS/sec
double_well	LSC-CP	double_well_shell (scale 1.0)	4	45772.913
double_well	raw CP	double_well_shell (scale 1.0)	4	88278.867
double_well	local	none	4	2.61e+05

Table 10: Generated metrics for four well graph. Means and standard errors are computed from raw CSV rows.

Target	Method	Bank	n	basin KL	mode/basin-TV	ESS/sec	ESS/grad
four_well	BAOAB	none	3	1.122	0.585	549.506	0.097
four_well	HMC	none	3	0.028	0.102	114.296	0.017
four_well	LSC-CP	minima_complete_graph (scale 1.0)	3	0.002	0.027	296.603	0.433
four_well	LSC-CP	minima_edge_graph (scale 1.0)	3	0.003	0.032	220.053	0.318
four_well	LSC-CP	random_matched_length_control	3	0.018	0.081	511.978	0.740
four_well	MALA	none	3	0.275	0.312	182.343	0.049
four_well	PT	none	3	0.004	0.035	3201.332	0.264
four_well	raw CP	minima_complete_graph (scale 1.0)	3	0.002	0.026	776.539	0.472
four_well	raw CP	minima_edge_graph (scale 1.0)	3	0.003	0.030	538.809	0.327
four_well	raw CP	random_matched_length_control	3	0.185	0.257	1179.940	0.716
four_well	local	none	3	0.275	0.305	532.329	0.099

Table 11: Generated metrics for manywell scaling. Means and standard errors are computed from raw CSV rows.

Target	Method	Bank	n	block KL	count KL	deep-count mean	ESS/sec	Levy evals
manywell_d16	BAOAB	none	3	3.605	4.874	0.075	1335.609	0.000
manywell_d16	HMC	none	3	2.938	4.874	0.156	266.894	0.000
manywell_d16	LSC-CP	manywell_block_flip	3	0.002	0.015	6.865	562.973	2.95e+08
manywell_d16	MALA	none	3	3.608	4.874	0.076	424.720	0.000
manywell_d16	PT	none	3	0.431	3.269	3.161	331.966	0.000
manywell_d16	raw CP	manywell_block_flip	3	0.254	2.067	4.044	624.517	0.000
manywell_d16	local	none	3	1.448	4.810	0.891	1272.990	0.000
manywell_d32	BAOAB	none	3	3.585	4.509	0.153	1358.250	0.000
manywell_d32	HMC	none	3	2.910	4.509	0.324	269.894	0.000
manywell_d32	LSC-CP	manywell_block_flip	3	0.003	0.056	13.885	476.763	5.90e+08
manywell_d32	MALA	none	3	3.761	4.509	0.126	413.177	0.000
manywell_d32	PT	none	3	0.505	4.112	5.733	335.205	0.000
manywell_d32	raw CP	manywell_block_flip	3	0.255	3.075	8.076	649.638	0.000
manywell_d32	local	none	3	1.232	4.509	2.303	1290.182	0.000
manywell_d64	BAOAB	none	3	3.822	4.175	0.309	1534.993	0.000
manywell_d64	HMC	none	3	2.932	4.175	0.633	294.039	0.000
manywell_d64	LSC-CP	manywell_block_flip	3	0.008	0.238	28.286	447.920	1.18e+09
manywell_d64	MALA	none	3	4.030	4.175	0.251	447.540	0.000
manywell_d64	PT	none	3	0.573	4.141	10.488	358.454	0.000
manywell_d64	raw CP	manywell_block_flip	3	0.242	3.877	16.462	540.720	0.000
manywell_d64	local	none	3	0.925	4.175	6.714	1413.954	0.000
manywell_d8	BAOAB	none	3	4.508	5.198	0.043	1467.754	0.000
manywell_d8	HMC	none	3	2.867	5.128	0.083	281.569	0.000
manywell_d8	LSC-CP	manywell_block_flip	3	8.58e-04	0.004	3.399	614.640	1.47e+08
manywell_d8	MALA	none	3	3.915	5.179	0.029	468.775	0.000
manywell_d8	PT	none	3	0.369	1.551	1.719	342.175	0.000
manywell_d8	raw CP	manywell_block_flip	3	0.258	1.008	2.010	710.733	0.000
manywell_d8	local	none	3	1.616	4.601	0.363	1323.734	0.000

Table 12: Generated metrics for muller10d basin communication. Means and standard errors are computed from raw CSV rows.

Target	Method	Bank	n	basin-TV	coverage time	coverage frac	ESS/sec
muller10d	LSC-CP	minima_complete_graph (scale 1.0)	4	0.012	1.911	0.636	5985.218
muller10d	LSC-CP	random_matched_length_control	4	0.124	2.367	0.152	1774.847
muller10d	raw CP	minima_complete_graph (scale 1.0)	4	0.181	1.818	0.697	4645.951
muller10d	raw CP	random_matched_length_control	4	0.030	2.274	0.296	3490.095
muller10d	local	none	4	0.110	2.410	0.155	5901.819

Table 13: Generated metrics for muller10d benchmark. Means and standard errors are computed from raw CSV rows.

Target	Method	Bank	n	basin-TV	ESS/sec	ESS/grad
muller10d	BAOAB	none	4	0.219	7245.638	5.266
muller10d	HMC	none	4	0.067	1353.333	0.654
muller10d	LSC-CP	minima_complete_graph (scale 1.0)	4	0.012	5937.556	13.997
muller10d	MALA	none	4	0.156	2598.186	2.133
muller10d	PT	none	4	0.100	2546.191	0.592
muller10d	raw CP	minima_complete_graph (scale 1.0)	4	0.181	4485.277	5.254
muller10d	local	none	4	0.110	5635.440	4.238

Table 14: Generated metrics for muller10d invariance. Means and standard errors are computed from raw CSV rows.

Target	Method	Bank	n	basin-TV
muller10d	LSC-CP	minima_complete_graph (scale 1.0)	4	0.012
muller10d	raw CP	minima_complete_graph (scale 1.0)	4	0.191
muller10d	local	none	4	0.019

Table 15: Generated metrics for triple well benchmark. Means and standard errors are computed from raw CSV rows.

Target	Method	Bank	n	mode/basin-TV	ESS/sec	ESS/grad
triple_well	BAOAB	none	4	0.266	1631.041	0.490
triple_well	HMC	none	4	0.009	685.008	0.223
triple_well	LSC-CP	adjacent (scale 1.0)	4	0.047	278.960	0.521
triple_well	MALA	none	4	0.007	1548.820	0.771
triple_well	PT	none	4	0.018	68096.255	9.246
triple_well	raw CP	adjacent (scale 1.0)	4	0.276	613.699	0.503
triple_well	local	none	4	0.739	2687.622	0.814

Table 16: Generated metrics for triple well invariance. Means and standard errors are computed from raw CSV rows.

Target	Method	Bank	n	basin-TV	CDF-sup
triple_well	LSC-CP	adjacent (scale 1.0)	4	0.011	0.019
triple_well	raw CP	adjacent (scale 1.0)	4	0.165	0.188
triple_well	local	none	4	0.009	0.016

Table 17: Generated metrics for triple well support. Means and standard errors are computed from raw CSV rows.

Target	Method	Bank	n	mode/basin-TV	middle-mass err	coverage frac	ESS/sec
triple_well	LSC-CP	adjacent (scale 1.0)	4	0.047	0.011	0.573	282.853
triple_well	LSC-CP	overlong (scale 1.0)	4	0.015	0.017	0.864	707.074
triple_well	LSC-CP	short (scale 0.4)	4	0.375	0.075	0.074	268.894
triple_well	raw CP	adjacent (scale 1.0)	4	0.276	0.170	0.368	706.769
triple_well	raw CP	overlong (scale 1.0)	4	0.168	0.163	0.445	742.813
triple_well	raw CP	short (scale 0.4)	4	0.454	0.151	0.068	699.504
triple_well	local	none	4	0.739	0.394	0.000	2784.303